Tool assembly for the end-2-end automation of controlled-access human data sharing

Data Stewardship Provenance and Compliance Kit (DS-PACK)

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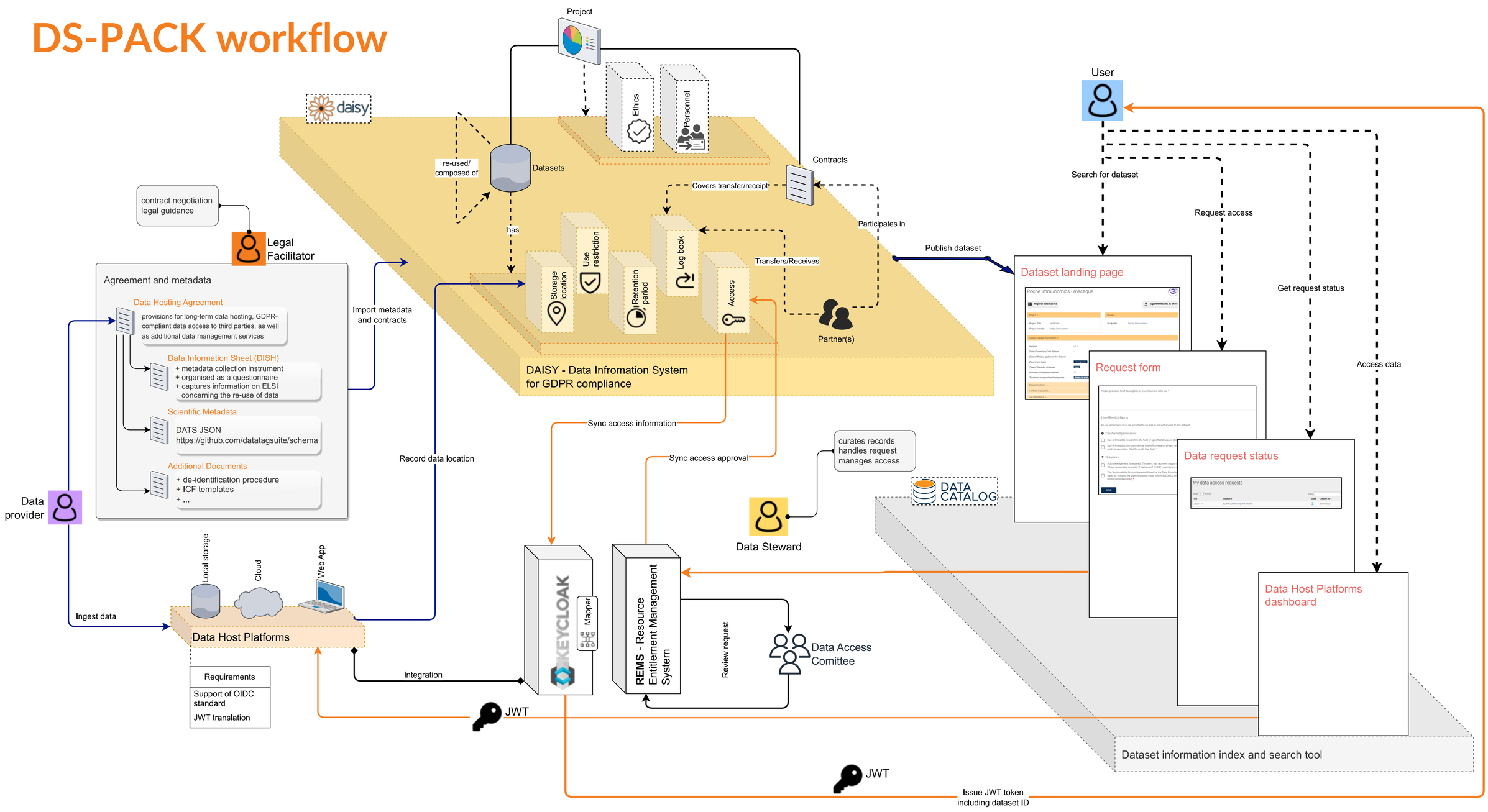
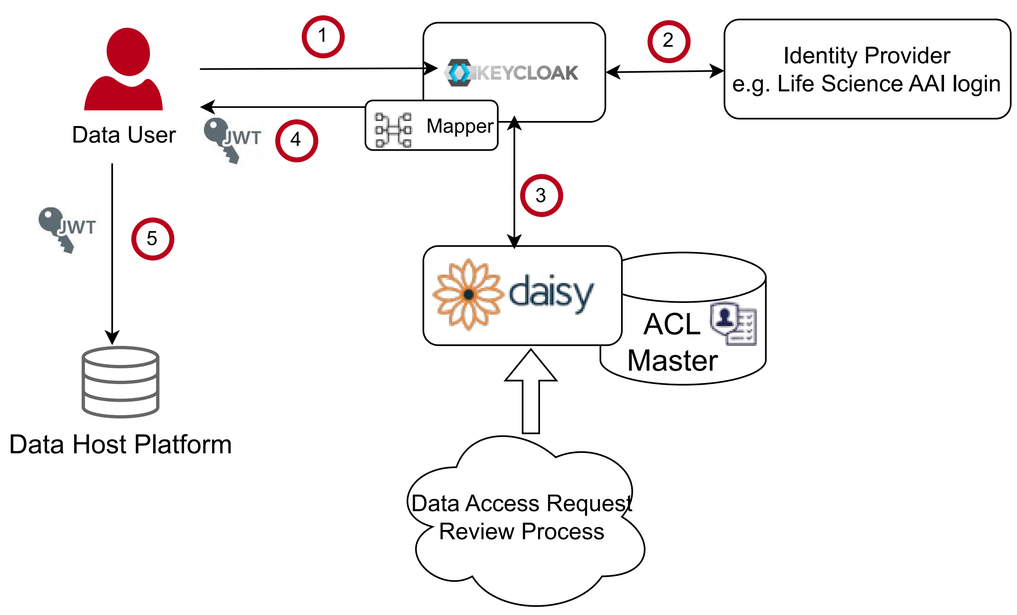
Centralised "controlled-access" genome/phenome archives have been the de facto means for sharing sensitive human data. The GDPR has diminished the choice of centralised repositories as it requires cross-border data safeguards, which are too costly or impossible to meet. In response, "repository federations" are being developed, where data discovery and access request management functions remain centralised; meanwhile, data storage and delivery are handled by distributed, often national, nodes. The federated approach succeeds in keeping data within national borders; however, it still requires the distributed nodes to abide by a uniform technology stack and data host platform. This limits the adoption of various database/warehouse solutions in clinical and translational research.

DS-PACK workflow



Authentication and Authorisation in DS-PACK



System Features

- Semi-automates the controlled access processes by outlining clear roles for data support teams which can relieve pressure from DACs and help scale up the sharing activities.
- Provides a solution for the long-term sustainability of clinical and translational data and associated metadata.
- Open-source and reusable component assembly
- Ensures GDPR compliance by design
- Acts as middleware translating the information from the data access request management system to commonly-used authentication and authorisation protocols, thereby allowing the sharing of data via any hosting platform supporting these protocols.
- Solves a problem inherent in distributed and federated data sharing, which is the centralisation of access control over distributed, heterogeneous hosting platforms.
- Supports the sharing of diverse biomedical data types and data delivery in diverse modalities.

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